ECE 340: Semiconductor Devices

Lecture 14: Generation-Recombination, Quasi-Fermi level

Prof. Hyunseok Kim

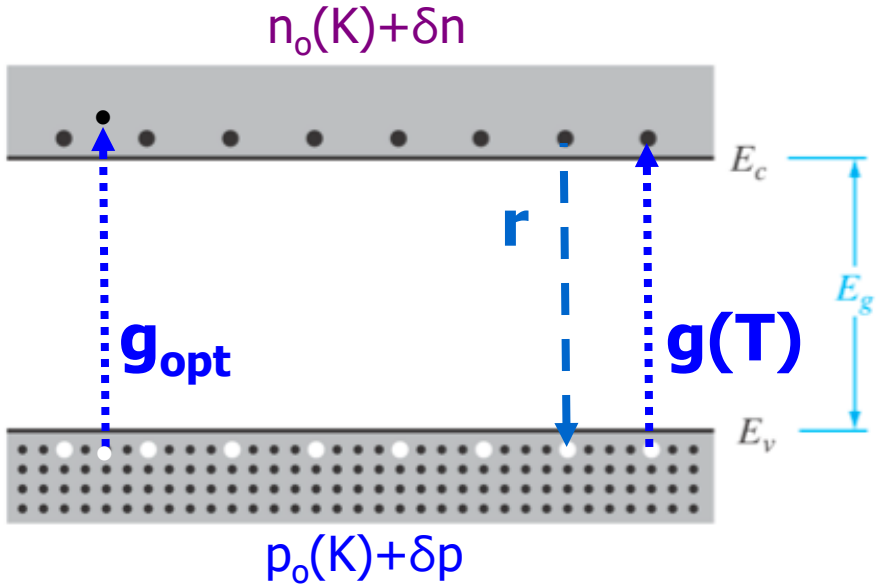
Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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(Recap) Steady State Generation

$$g = g(T) + g_{opt} = r = \alpha_r np$$
$$= \alpha_r (n_0 + \delta n)(p_0 + \delta p)$$
$$= \alpha_r \left[n_0 p_0 + n_0 \delta p + p_0 \delta n + \delta n \delta p \right]$$
$$= \alpha_r \left[n_0 p_0 + (n_0 + p_0)\delta n + \delta n^2 \right]$$
$$= \alpha_r n_0 p_0 + \alpha_r \left[(n_0 + p_0)\delta n + \delta n^2 \right]$$



Recall: $g(T) = \alpha_r n_i^2 = \alpha_r n_0 p_0$

so

$$g_{op} = \alpha_r \left[(n_0 + p_0)\delta n + \delta n^2 \right] \approx \alpha_r (n_0 + p_0)\delta n = \frac{\delta n}{\tau_n}$$

$$\tau_n = \frac{1}{\alpha_r (n_0 + p_0)}$$

and

$$\delta n = \delta p = g_{op} \tau_n$$

Assume δn is small compared to majority carrier concentration

(Recap) Solving the Differential Equation

p-type Material

1) Assume low-level injection:

neglect $\delta n^2(t)$

2) p-type material so $p_0 >> n_0$

$$\frac{d\delta n(t)}{dt} = -\alpha_r\left[\left(n_0 + p_0\right)\delta n(t) + \delta n^2(t)\right]$$

$$\approx -\alpha_r\left(n_0 + p_0\right)\delta n(t) \approx -\alpha_r p_0 \delta n(t)$$

The solution is:

$$\delta n(t) = \Delta n e^{-\alpha_r p_0 t} = \Delta n e^{-t/\tau_n}$$

where

$$\Delta n = \delta n(t = 0)$$

$$\tau_n = \frac{1}{\alpha_r p_0} \quad \text{or}$$

$$\tau_n = \frac{1}{\alpha_r\left(n_0 + p_0\right)}$$

n-type Material

1) Assume low-level injection:

neglect $\delta p^2(t)$

2) n-type material so $n_0 >> p_0$

$$\frac{d\delta p(t)}{dt} = -\alpha_r\left[\left(n_0 + p_0\right)\delta p(t) + \delta p^2(t)\right]$$

$$\approx -\alpha_r\left(n_0 + p_0\right)\delta p(t) \approx -\alpha_r n_0 \delta p(t)$$

The solution is:

$$\delta p(t) = \Delta p e^{-\alpha_r n_0 t} = \Delta p e^{-t/\tau_p}$$

where

$$\Delta p = \delta p(t = 0)$$

$$\tau_p = \frac{1}{\alpha_r n_0} \quad \text{or}$$

$$\tau_p = \frac{1}{\alpha_r\left(n_0 + p_0\right)}$$

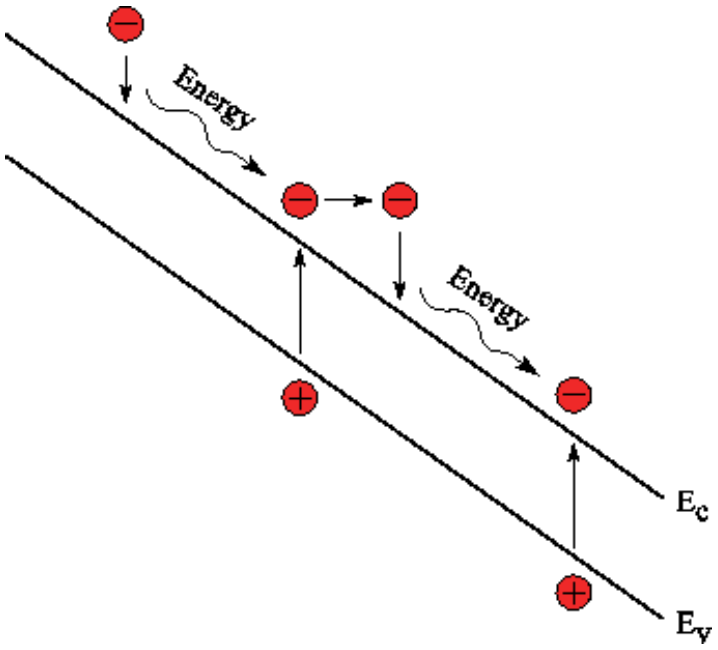
(Recap) Important Definition

- ***Minority Carrier Lifetime*** or ***Recombination Lifetime***: the time constant by which the excess carrier population decays to the equilibrium value

$$\tau_n = \frac{1}{\alpha_r \left(n_0 + p_0 \right)}$$

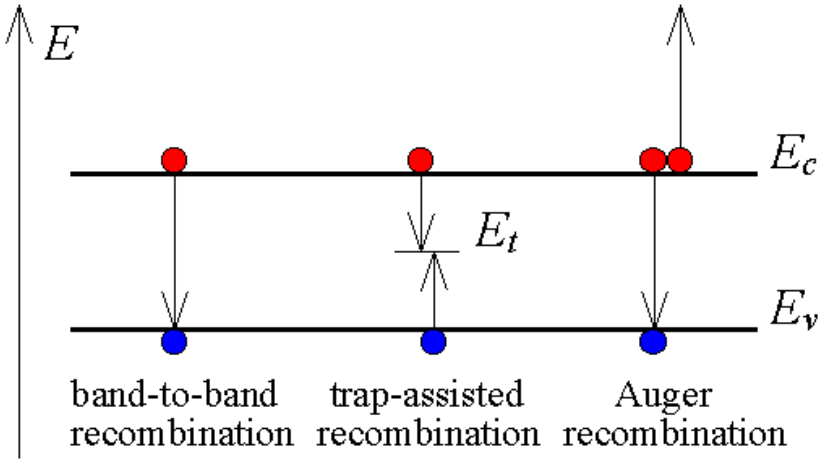
Other G-R Processes

Impact Ionization



http://www.iue.tuwien.ac.at/phd/entner/node11.html

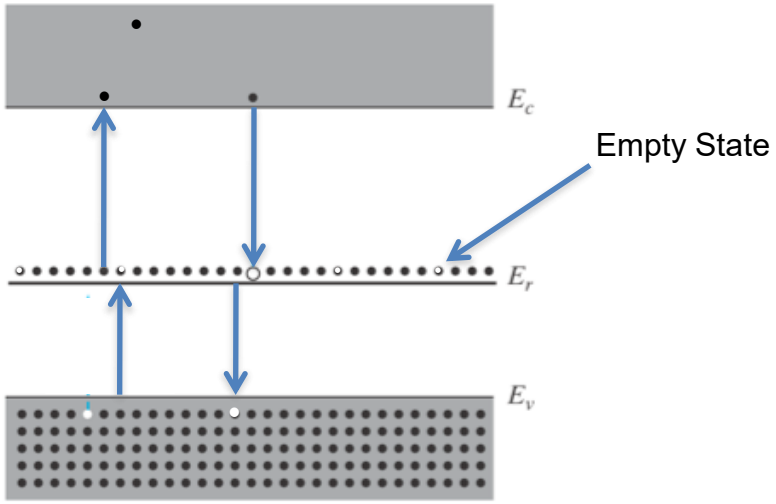
Auger, SRH GR



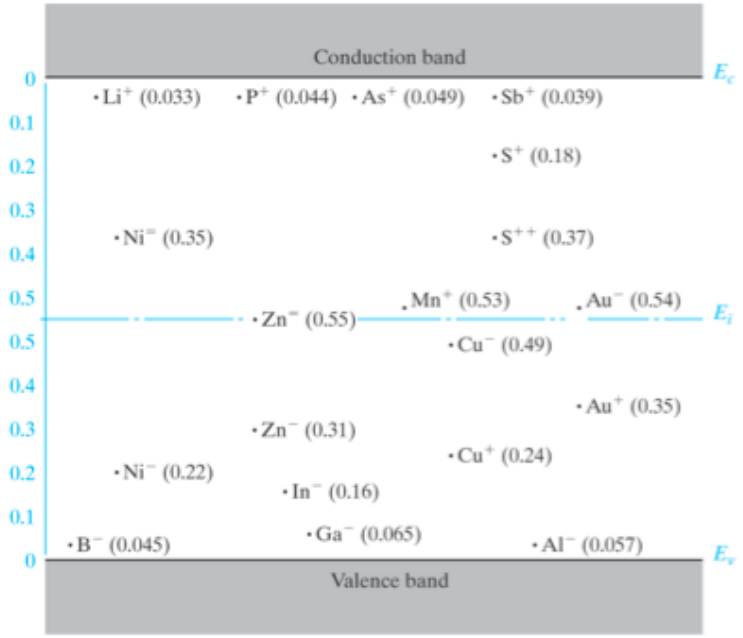
http://ecee.colorado.edu/~bart/ecen5355/newbook/chapter2/recomb.htm

G-R Through Trapping Centers (SRH)

Generation-Recombination



Traps in Silicon



From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.
ISBN 0-13-149726-X. © 2006 Pearson Education, Inc., Upper Saddle River, NJ. All rights reserved.

Quasi Fermi Levels

Steady Illumination: Revisited

$$r = \alpha_r np = \alpha_r (n_o + \delta n)(p_o + \delta p)$$
$$= \alpha_r [n_o p_o + n_o \delta p + p_o \delta n + \delta n \delta p]$$
$$= \alpha_r [n_o p_o + (n_o + p_o)\delta n + \delta n^2]$$
$$= g(T) + g_{opt}$$

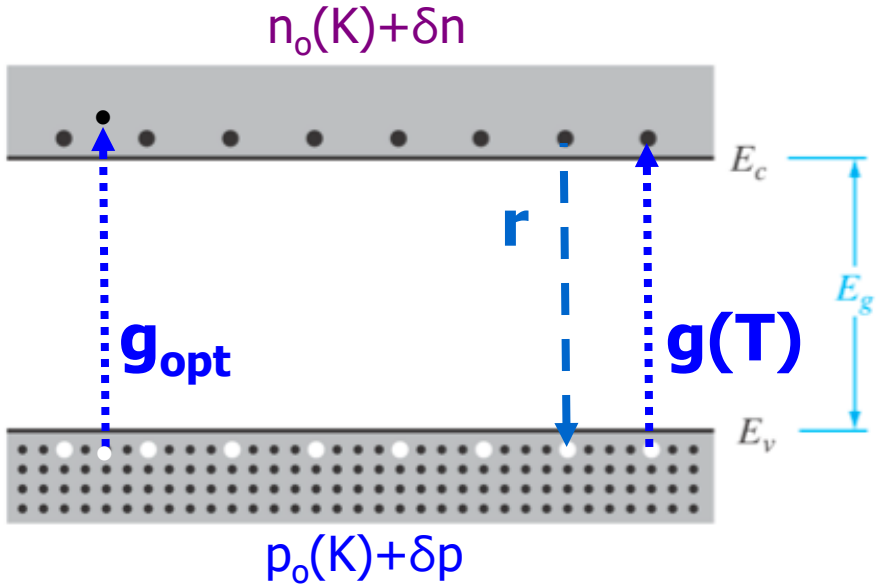
Recall: $g(T) = \alpha_r n_i^2 = \alpha_r n_0 p_0$

so

$$g_{op} = \alpha_r (n_0 + p_0)\delta n = \frac{\delta n}{\tau_n}$$

and

$$\delta n = \delta p = g_{op} \tau_n$$



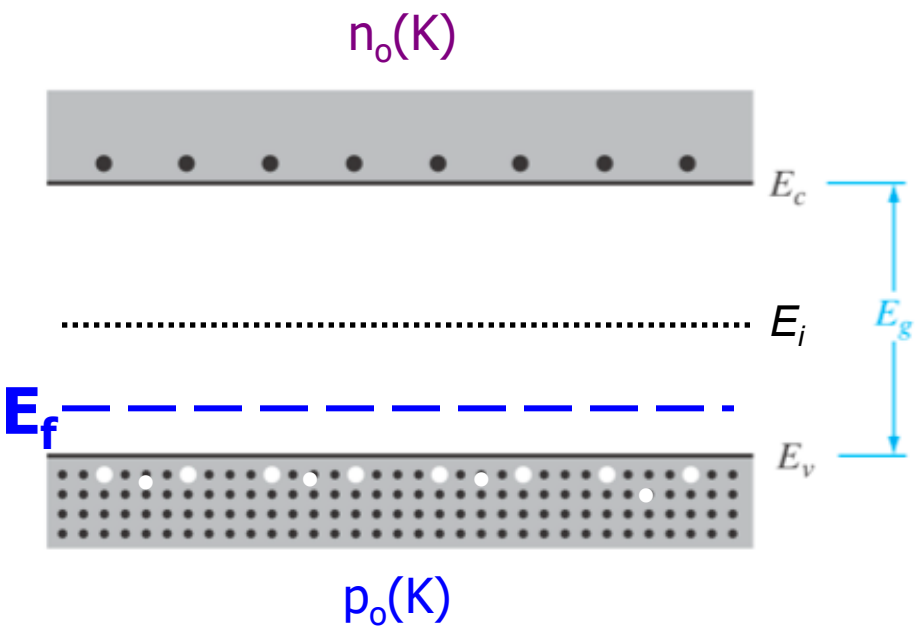
What Happens to the Fermi Level?

Quasi-Fermi Level

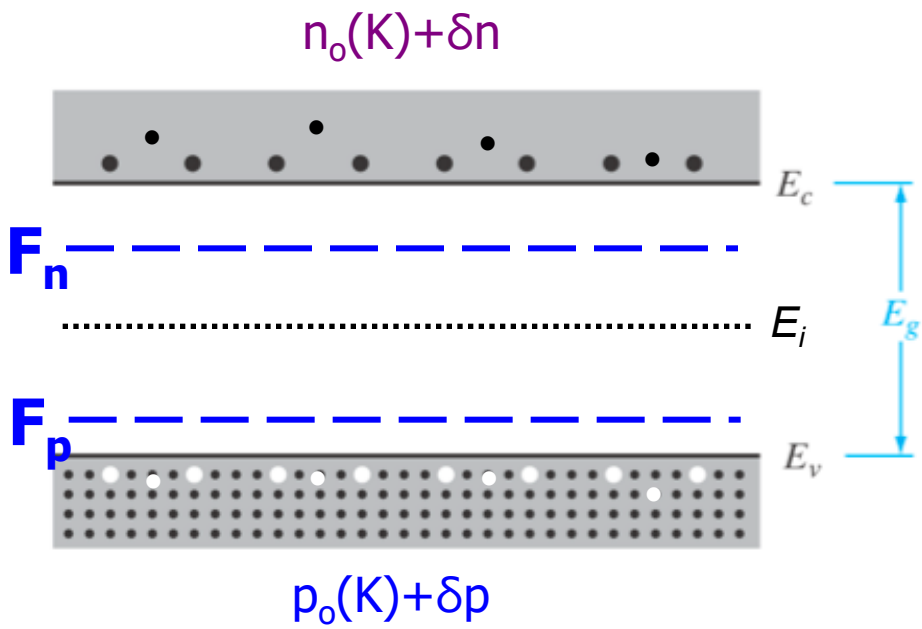
- A variety of mechanisms can create excess carriers (electrons and holes)
- The carriers may or may not be created in equal numbers
  - This depends upon the mechanism that created them
- Regardless of how they were created, the quantity of excess carriers can be described by integrating the density of states times the probability of occupancy
- The probability of occupancy is still a Fermi distribution, but with a modified Fermi energy, the “Quasi-Fermi Energy” or “Quasi-Fermi Level”

What Happens to the Fermi Level?

Equilibrium



With Excess Carriers



The Fermi level in the presence of excess carriers is called the "quasi-Fermi level"

$$n_0 = n_i e^{(E_F - E_i)/kT}$$

$$p_0 = n_i e^{(E_i - E_F)/kT}$$

$$n = n_i e^{(F_n - E_i)/kT}$$

$$p = n_i e^{(E_i - F_p)/kT}$$

Quasi Fermi Levels

- Electron Quasi-Fermi Level:

$$n = n_i e^{(F_n - E_i)/kT} = n_i e^{(F_n - E_i + E_f - E_f)/kT}$$

$$= n_i e^{(E_f - E_i)/kT} e^{(F_n - E_f)/kT} = n_o e^{(F_n - E_f)/kT}$$

$$\delta n = n - n_o = n_o (e^{(F_n - E_f)/kT} - 1) = g_{op} \tau_n$$

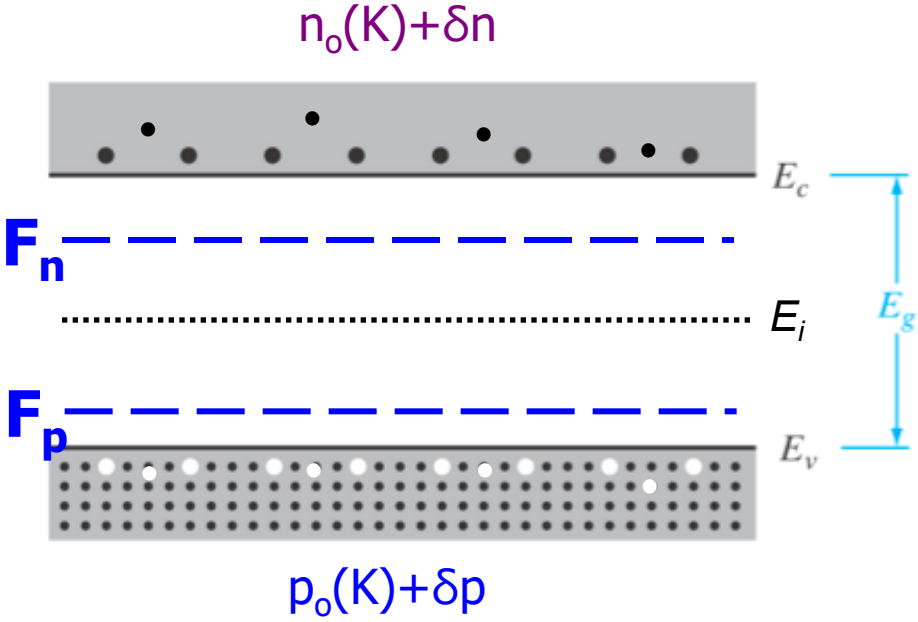
- Hole Quasi-Fermi Level:

$$p = p_o e^{(E_f - F_p)/kT}$$

$$\delta p = p - p_o = p_o (e^{(E_f - F_p)/kT} - 1) = g_{op} \tau_p$$

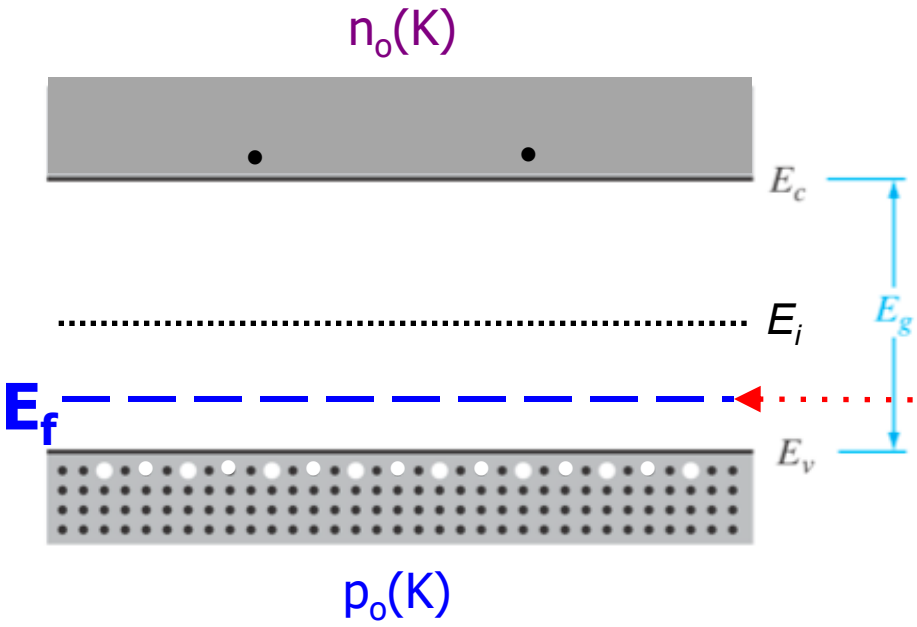
$$\delta n = \delta p = g_{op} \tau_n$$

Band Diagram, F_n and F_p

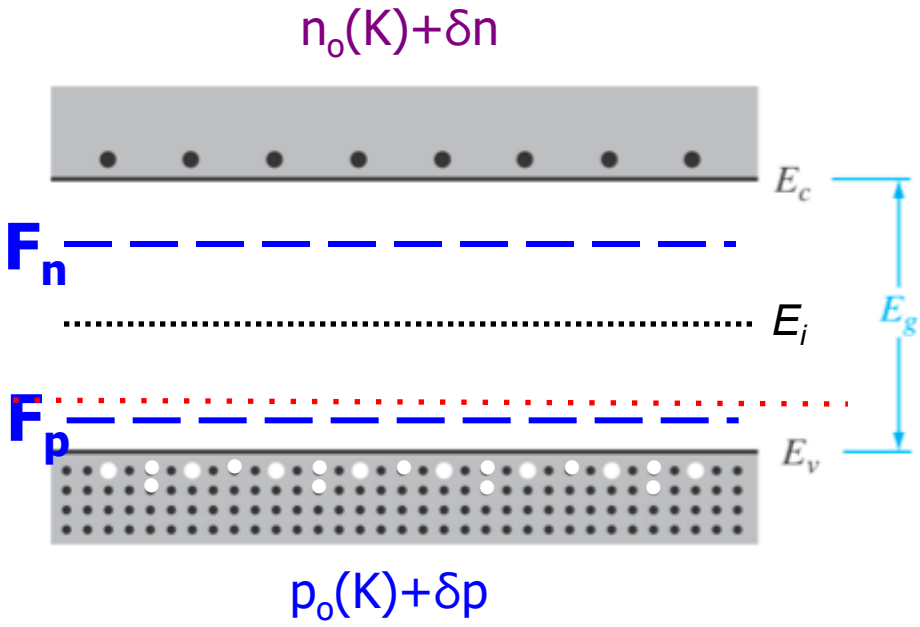


Quasi-Fermi Levels in p-type Material

Equilibrium



With Excess Carriers

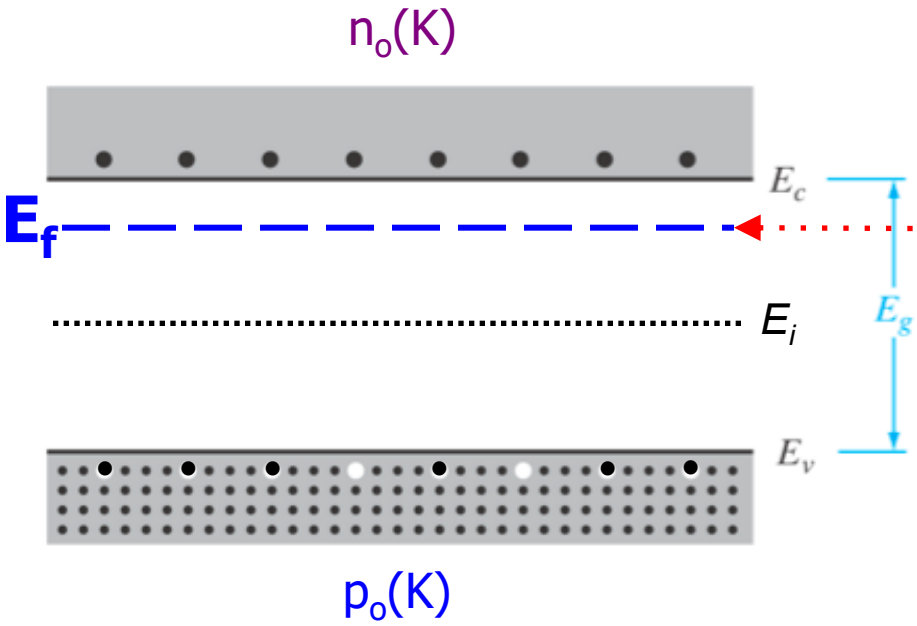


$$n = n_i e^{(F_n - E_i)/kT}$$

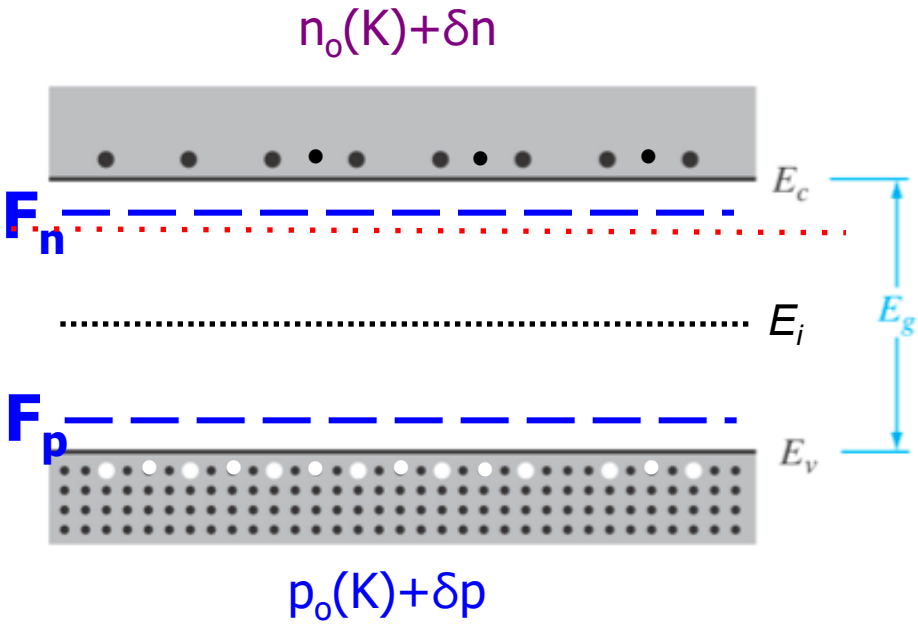
$$F_n - E_i = kT \ln\left(\frac{n}{n_i}\right)$$

Quasi-Fermi Levels in n-type Material

Equilibrium



With Excess Carriers



$$p = n_i e^{(E_i - F_p)/kT}$$

$$E_i - F_p = kT \ln\left(\frac{p}{n_i}\right)$$

Example: Steady-State Generation in Si

1) 10^{13} EHP/cm^3 optically created per microsecond

Therefore, $g_{opt} = \dfrac{10^{13}\,\text{EHP/cm}^3}{10^{-6}\,s} = 10^{19}$ EHP/cm^{3}s

2) $n_o = 10^{14}$ cm^{-3} in equilibrium (no illumination)

Therefore, $p_o = \dfrac{n_i^2}{n_o} = \dfrac{2.25 \times 10^{20}}{10^{14}}\text{cm}^{-3} = 2.25 \times 10^6$

3) The equilibrium Fermi level (no illumination) is calculated:

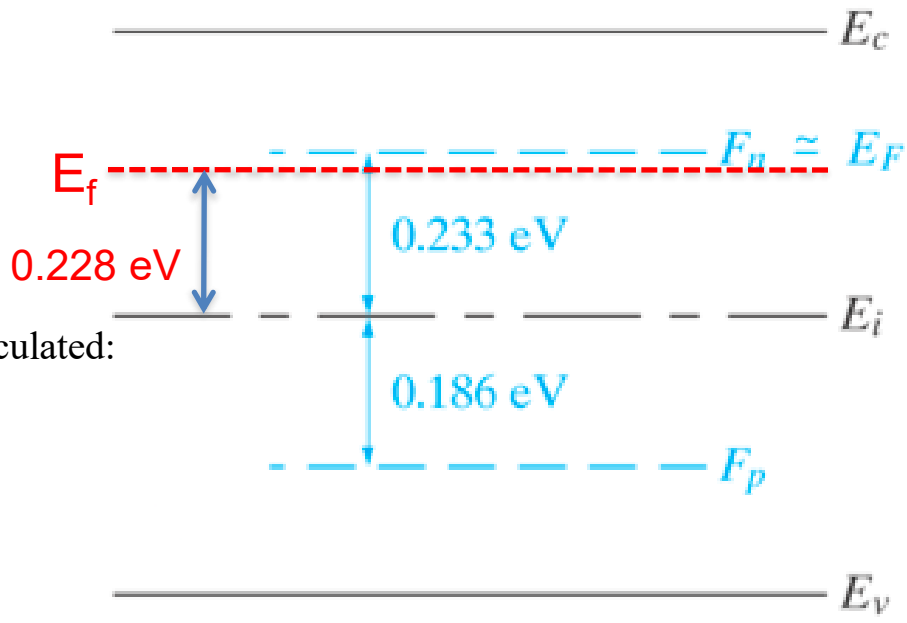
$n_o = 10^{14} = n_i e^{(E_f - E_i)/kT} = (1.5 \times 10^{10}) e^{(E_f - E_i)/kT}$

$E_f - E_i = 0.0259 \ln(6.7 \times 10^3) = 0.228 eV$

4) $\tau_n = \tau_p = 2\mu\text{sec}$

$g_{opt} = \dfrac{\delta n}{\tau}$

Therefore $\delta n = \delta p = g_{opt}\tau_n = 2 \times 10^{13}$



5) $n = n_o + \delta n = 1.2 \times 10^{14}\,\text{cm}^{-3} = (1.5 \times 10^{10}) e^{(F_n - E_i)/kT}$, kT=0.0259 eV

Therefore $F_n - E_i = 0.0259 \ln(8 \times 10^3) = 0.233 eV$

6) $p = p_o + \delta p = 2 \times 10^{13}\,\text{cm}^{-3} = (1.5 \times 10^{10}) e^{(E_i - F_p)/kT}$

Therefore $E_i - F_p = 0.0259 \ln(1.3 \times 10^3) = 0.186 eV$

Note: $np \neq n_i^2$ when excess carriers are present!

Fermi Level Comments

- The separation of the electron and hole quasi Fermi levels (F_n-F_p) provides a metric for the deviation from equilibrium
- The individual deviation F_n and F_p from E_F is a measure of how far the electron and hole populations are from equilibrium
- Depicting the quasi Fermi Levels on a band diagram gives a good representation of how a sample is locally deviating from equilibrium

$$np = \left(n_i e^{(F_n - E_i)/kT}\right)\left(n_i e^{(E_i - F_p)/kT}\right)$$

$$= n_i^2 e^{(F_n - E_i)/kT} e^{(E_i - F_p)/kT}$$

$$= n_i^2 e^{(F_n - E_i + E_i - F_p)/kT}$$

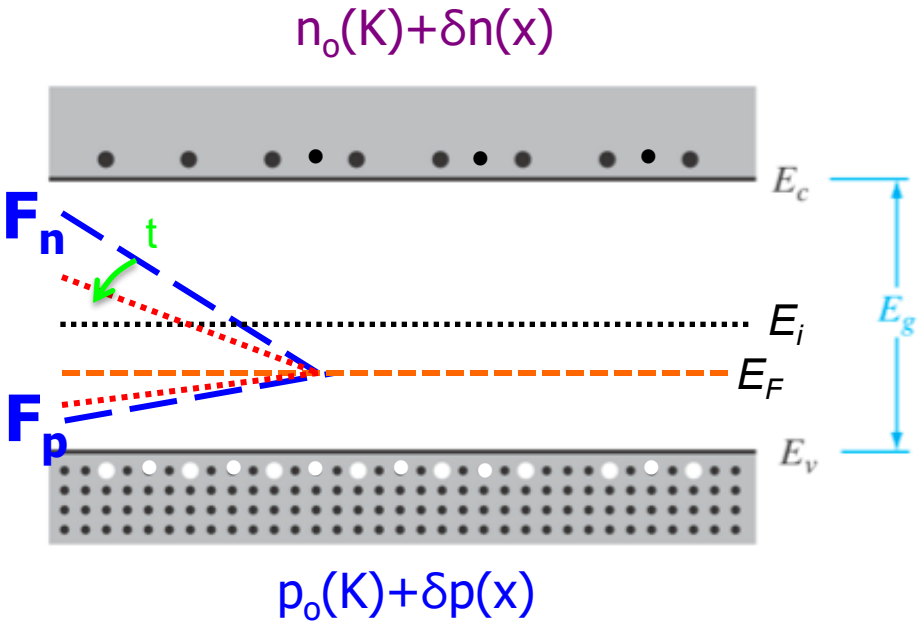
$$= n_i^2 e^{(F_n - F_p)/kT}$$



Valid for carrier densities such that quasi-Fermi level is 3kT away from band edge

Position and Time Dependence

- The quasi Fermi level can depend upon position (x)
- The quasi Fermi level will generally depend upon time (t) unless the device or material is under steady-state conditions

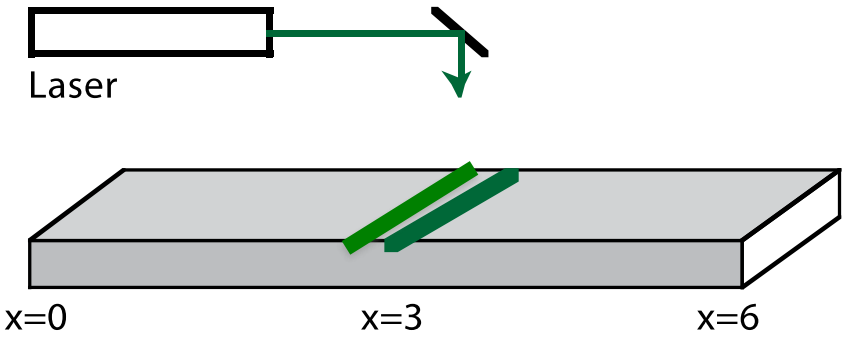


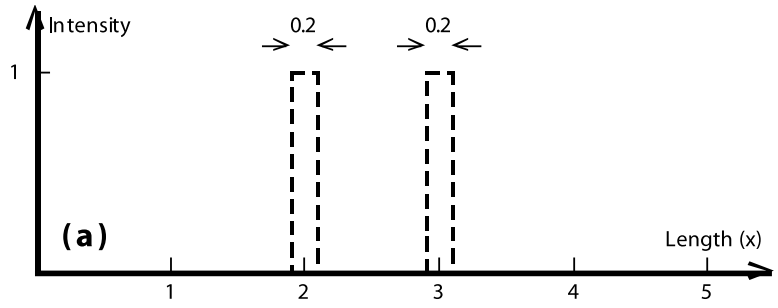
$$n = n_i e^{\left(F_n(x,t)-E_i\right)/kT}$$

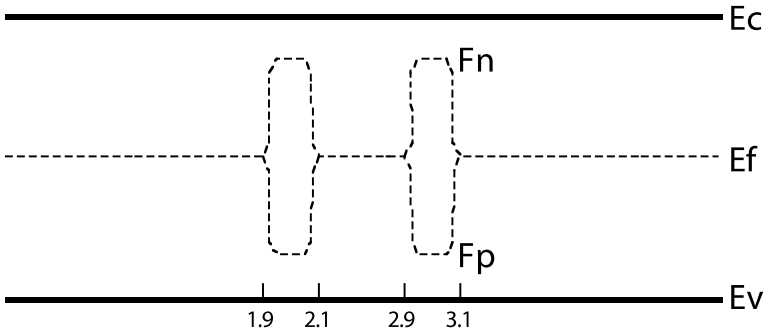
$$p = n_i e^{\left(E_i-F_p(x,t)\right)/kT}$$

F_n, F_p more than 3kT away from E_c, E_v

Instantaneous Pulse







At $t=0^+$ (Neglecting Diffusion)

Photoconductivity

Photoconductivity

Conductivity:

$$\sigma = q(n\mu_n + p\mu_p) = q(n_o + \delta n)\mu_n + q(p_o + \delta p)\mu_p$$

$$= q(n_o\mu_n + p_o\mu_p) + q(\delta n\mu_n + \delta p\mu_p)$$

$$= \sigma_o + \Delta\sigma$$

$n = n_o + \delta n$

$p = p_o + \delta p$

Photoconductivity Change:

$$\Delta\sigma = q(\delta n\mu_n + \delta p\mu_p) = qg_{opt}(\tau_n\mu_n + \tau_p\mu_p)$$

Since $\delta n = g_{opt}\tau_n$ and $\delta p = g_{opt}\tau_p$

- Maximizing Δσ:
  - Large mobility
  - Long carrier lifetime